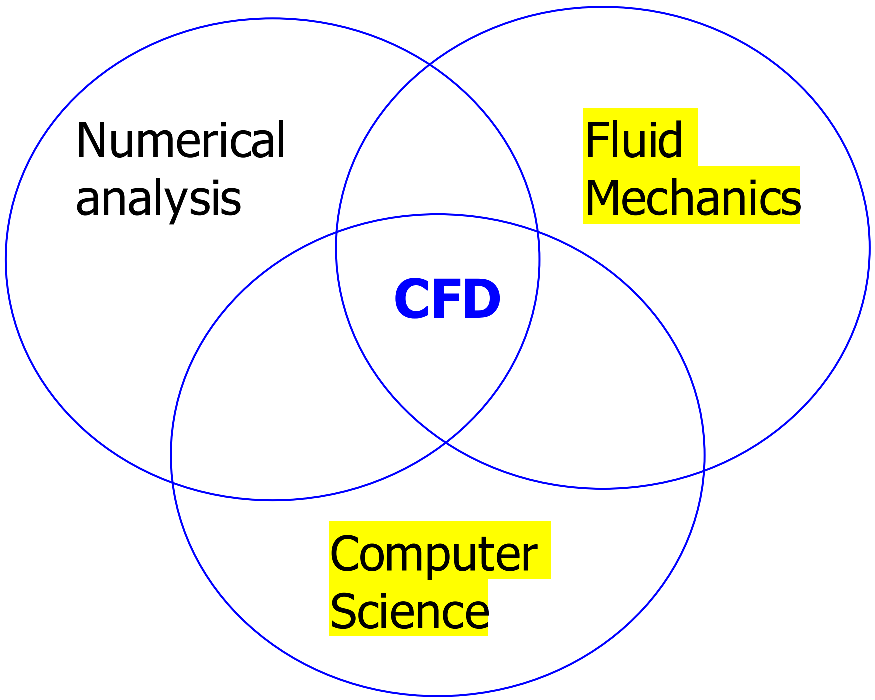
Overview of fluid mechanics governing equations

Boundary conditions at fluid-solid interface

A Fortran demo code



Basic theoretical knowledge

Fundamental theory

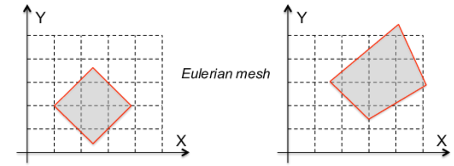
Lagrangian and Eulerian specification of the flow field

In the *Eulerian specification of a field*, it is represented as *a function of position x and time* t. For example, the flow velocity is represented by a function

$B(x, y, z, t)$; $\mathrm{x(t), y(t), z(t)}$ following a fluid element

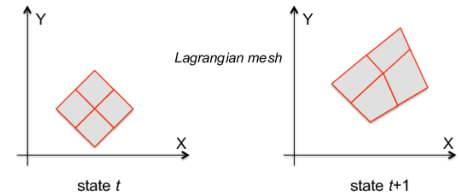
$$\Rightarrow \frac{DB}{Dt} = \frac{\partial B}{\partial t} + \frac{\partial B}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial B}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial B}{\partial z}\frac{\partial z}{\partial t} = \frac{\partial B}{\partial t} + u\frac{\partial B}{\partial x} + v\frac{\partial B}{\partial y} + w\frac{\partial B}{\partial z}$$

$$= \frac{\partial B}{\partial t} + \vec{u} \cdot \nabla B$$



In the *Lagrangian specification*, individual fluid parcels are followed through time. In the Lagrangian description, the flow is described by a function $B(x_0, y_0, z_0, t)$

$$\Rightarrow \frac{dB}{dt} = \frac{\partial B(x_0, y_0, z_0, t)}{\partial t}$$



Basic theoretical knowledge

Fundamental theory

Claude-Louis Navier (1785-1836)

George Stokes (1819-1903)

Incompressible Navier–Stokes equations (convective form, no body force)

$$\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\frac{1}{\rho}\nabla p + \boldsymbol{g} + \frac{\mu}{\rho}\nabla^2 \boldsymbol{u}, \qquad \boldsymbol{u} = (v_x, v_y, v_z)$$

With body force

x direction

$$\rho\left(\frac{\partial v_x}{\partial t} + v_x\frac{\partial v_x}{\partial x} + v_y\frac{\partial v_x}{\partial y} + v_z\frac{\partial v_x}{\partial z}\right) = -\frac{\partial p}{\partial x} + \rho g_x + \mu\left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}\right)$$

y direction

$$\rho\left(\frac{\partial v_y}{\partial t} + v_x\frac{\partial v_y}{\partial x} + v_y\frac{\partial v_y}{\partial y} + v_z\frac{\partial v_y}{\partial z}\right) = -\frac{\partial p}{\partial y} + \rho g_y + \mu\left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}\right)$$

z direction

$$\rho\left(\frac{\partial v_z}{\partial t} + v_x\frac{\partial v_z}{\partial x} + v_y\frac{\partial v_z}{\partial y} + v_z\frac{\partial v_z}{\partial z}\right) = -\frac{\partial p}{\partial z} + \rho g_z + \mu\left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}\right)$$

Stokes: Setting up the differential equation for a viscous fluid at low Reynolds number.

从牛顿第二定律到Navier-Stokes方程

$$\sum \boldsymbol{F} = \text{表面力} + \text{体力} = m\boldsymbol{a} = \frac{d(m\boldsymbol{u})}{dt}$$

$\rho =$ 密度

$u =$ 速度

$\boldsymbol{\tau} =$ 应力 （单位面积受力）

流体微团→空间固定控制体

$$\frac{D(m\boldsymbol{u})}{Dt} = \frac{\partial(m\boldsymbol{u})}{\partial t} + \frac{\partial(m\boldsymbol{u})}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial(m\boldsymbol{u})}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial(m\boldsymbol{u})}{\partial z}\frac{\partial z}{\partial t}$$

$$= \frac{\partial(m\boldsymbol{u})}{\partial t} + u_x\frac{\partial(m\boldsymbol{u})}{\partial x} + u_y\frac{\partial(m\boldsymbol{u})}{\partial y} + u_z\frac{\partial(m\boldsymbol{u})}{\partial z}$$

$$= dV\left(\frac{\partial(\rho\boldsymbol{u})}{\partial t} + \boldsymbol{u}\cdot\nabla(\rho\boldsymbol{u})\right)$$

$$\sum \boldsymbol{F} = \oint \boldsymbol{\tau}\cdot\boldsymbol{n}\,dS + dV\rho\boldsymbol{g} = \int \nabla\cdot\boldsymbol{\tau}\,dV + dV\rho\boldsymbol{g}$$ 高斯定理

$$\frac{\partial(\rho\boldsymbol{u})}{\partial t} + \boldsymbol{u}\cdot\nabla(\rho\boldsymbol{u}) = \frac{D(\rho\boldsymbol{u})}{Dt} = \nabla\cdot\boldsymbol{\tau} + \rho\boldsymbol{g} \quad \leftarrow \quad \boldsymbol{\tau} = -\nabla p + \mu\left(\nabla\boldsymbol{u} + \nabla^{\mathrm{T}}\boldsymbol{u}\right)$$

The Navier-Stokes Equations

ρ – fluid density

U_j – fluid velocity

P – local pressure

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u_j)}{\partial x_j} = 0$$

$$\rho \frac{\partial u_i}{\partial t} + \rho u_j \frac{\partial u_i}{\partial x_j} = \rho g_i - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{D}\nabla \cdot \vec{u}\delta_{ij}\right) + \mu^V \nabla \cdot \vec{u}\delta_{ij}\right]$$

$\mu \equiv$ shear viscosity; $\mu^V \equiv$ bulk viscosity or volume viscosity

Fluid-dynamics stress due to fluid motion

Viscous stress

$$\sigma_{ij} = \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{D}\nabla \cdot \vec{u}\delta_{ij}\right) + \mu^V \nabla \cdot \vec{u}\delta_{ij} = 2\mu\left(S_{ij} - \frac{1}{D}\nabla \cdot \vec{u}\delta_{ij}\right) + \mu^V \nabla \cdot \vec{u}\delta_{ij}$$

$D \equiv$ spatial dimension; $S_{ij} \equiv \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$ Strain rate

Total stress: $\tau_{ij} \equiv -p\delta_{ij} + \sigma_{ij}$

Governing equation valid for both laminar and turbulent flows

Kundu, Cohen, Dowling, *Fluid Mechanics*

Energy equation (1st Law of Thermodynamics in differential form)

Primitive form (1), exactly as stated in the First Law of Thermodynamics

e – internal energy

$$\frac{\partial}{\partial t}\left[\rho\left(e+\frac{1}{2}u^2-g_k x_k\right)\right]+\frac{\partial}{\partial x_j}\left[\rho u_j\left(e+\frac{1}{2}u^2-g_k x_k\right)\right]=\frac{\partial}{\partial x_i}\left[\left(-p\delta_{ij}+\sigma_{ij}\right)u_j\right]+\frac{\partial}{\partial x_j}\left[k\frac{\partial T}{\partial x_j}\right]$$

T – local temperature

Primitive form (2)

$$\frac{\partial}{\partial t}\left[\rho\left(e+\frac{1}{2}u^2\right)\right]+\frac{\partial}{\partial x_j}\left[\rho u_j\left(e+\frac{1}{2}u^2\right)\right]=\rho g_j u_j+\frac{\partial}{\partial x_i}\left[\left(-p\delta_{ij}+\sigma_{ij}\right)u_j\right]+\frac{\partial}{\partial x_j}\left[k\frac{\partial T}{\partial x_j}\right]$$

The Equation of State

$$p=f(\rho,T,\ldots)$$

Ex 1: ideal gas: $p=\rho RT$

Ex 2: The van der Waals Equation of State:

$$\left(p+a\rho^2\right)\left(\frac{1}{\rho}-b\right)=RT$$

Mechanical Energy equation

$$\frac{\partial}{\partial t}\left[\frac{1}{2}\rho u^2\right]+\frac{\partial}{\partial x_j}\left[\rho u_j\frac{1}{2}u^2\right]=\rho g_j u_j+u_i\frac{\partial}{\partial x_j}\left[-p\delta_{ij}+\sigma_{ij}\right]$$

Internal energy equation

$$\rho \frac{\partial e}{\partial t} + \rho u_j \frac{\partial e}{\partial x_j} = -p\nabla \cdot \vec{u} + 2\mu \left(S_{ij} - \frac{1}{3}\nabla \cdot \vec{u}\delta_{ij} \right)^2 + \mu^V (\nabla \cdot \vec{u})^2 + \frac{\partial}{\partial x_j}\left(k \frac{\partial T}{\partial x_j} \right)$$

Simplified Form (using dissipation rate ε):

$$\rho \frac{\partial e}{\partial t} + \rho u_j \frac{\partial e}{\partial x_j} = -p\nabla \cdot \vec{u} + \rho\varepsilon + \frac{\partial}{\partial x_j}\left(k \frac{\partial T}{\partial x_j} \right)$$

where the dissipation rate per unit mass (irreversible energy transfer)

$$\varepsilon = 2\frac{\mu}{\rho}\left(S_{ij} - \frac{1}{3}\nabla \cdot \vec{u}\delta_{ij} \right)^2 + \frac{\mu^V}{\rho}(\nabla \cdot \vec{u})^2$$

Entropy equation – the 2nd law of thermodynamics

$$\rho \frac{\partial e}{\partial t} + \rho u_j \frac{\partial e}{\partial x_j} = -p\nabla \cdot \vec{u} + \rho\varepsilon + \frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right)$$

Applying the Gibbs relation $de = Tds - pd(1/\rho)$

$$\frac{Ds}{Dt} = \frac{\varepsilon}{T} + \frac{1}{\rho T}\frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right)$$

$$\frac{Ds}{Dt} = \frac{1}{\rho}\frac{\partial}{\partial x_j}\left(\frac{k}{T}\frac{\partial T}{\partial x_j}\right) + \frac{k}{\rho T^2}(\nabla T)^2 + \frac{\varepsilon}{T}$$

Entropy transfer
due to heat transfer

Entropy production
due to finite temperature difference
& viscous dissipation

T: temperature 温度
p: pressure 压力
ρ: density 密度

e: specific internal energy 单位质量内能
ϵ: viscous dissipation 单位质量粘性耗散率
s: specific entropy 单位质量的熵

Transport coefficients:
μ: shear viscosity 剪切粘性系数
μ^V: bulk viscosity 体积粘性系数
k: conductivity 导热系数

Therefore, the Second Law of Thermodynamics simply requires that

$$\mu \geq 0$$
$$\mu^V \geq 0$$
$$k \geq 0$$

Viscosity of air as a function of temperature

Model 1: The Sutherland equation

$$\mu = \frac{bT^{1.5}}{T+S} \qquad where \quad b = 1.458\times10^{-6}\,\frac{kg}{m\cdot s\cdot K^{0.5}} \qquad S = 110.4\ K$$

Model 2: Power Law model

$$\mu = \mu_0\left(\frac{T}{T_0}\right)^{0.7} \qquad where \quad T_0 = 273K,$$

$$\mu_0 = \mu(273\ K) = 1.71\times10^{-5}\,\frac{kg}{m\cdot s}$$

A demo Fortran code: viscosity.f90

```
Program Viscosity
   real*4 mu1, mu2, b, S, mu0, T, TK, T0
   integer i

! Compare two models of viscosity

   b = 1.458e-6
   S = 110.4

   mu0 = 1.71e-5
   T0 = 273.

      do i = -5,10
      T = 10.0*real(i)   ! in C
! Convert to K
      TK = T + 273.
      mu1 = b*TK**1.5/(TK+S)
      mu2 = mu0*(TK/T0)**0.7
  write(10,'(2F7.1,2(1PE15.5))') T,TK,mu1,mu2
! Print out a data table
      end do

End Program Viscosity
```

Compile, run, and check the results:

```
gfortran -O3 viscosity.f90
./a.out
```

Results

```
 -50.0  223.0    1.45629E-05    1.48421E-05
 -40.0  233.0    1.51005E-05    1.53049E-05
 -30.0  243.0    1.56279E-05    1.57618E-05
 -20.0  253.0    1.61456E-05    1.62131E-05
 -10.0  263.0    1.66539E-05    1.66591E-05
   0.0  273.0    1.71534E-05    1.71000E-05
  10.0  283.0    1.76442E-05    1.75361E-05
  20.0  293.0    1.81269E-05    1.79676E-05
  30.0  303.0    1.86016E-05    1.83947E-05
  40.0  313.0    1.90688E-05    1.88176E-05
  50.0  323.0    1.95287E-05    1.92364E-05
  60.0  333.0    1.99815E-05    1.96514E-05
  70.0  343.0    2.04276E-05    2.00626E-05
  80.0  353.0    2.08672E-05    2.04703E-05
  90.0  363.0    2.13005E-05    2.08745E-05
 100.0  373.0    2.17277E-05    2.12754E-05
```

Low speed viscous flows

$$\frac{\partial \rho}{\partial t}+\frac{\partial\left(\rho u_j\right)}{\partial x_j}=0 \quad \rightarrow \quad \nabla\cdot\vec{u}+\underbrace{\frac{1}{\rho}\frac{D\rho}{Dt}}^{\sim O(Ma^2)}=0 \quad \rightarrow \quad \nabla\cdot\vec{u}=0$$

Zero divergence also greatly simplifies the momentum equation

$$\rho\frac{\partial u_i}{\partial t}+\rho u_j\frac{\partial(u_i)}{\partial x_j}=-\rho g e_{i3}-\frac{\partial p}{\partial x_i}+\mu\frac{\partial^2 u_i}{\partial x_j\partial x_j}$$

The nonlinear term

The viscous term

Boussinesq approximation

$p_m=p+\rho_0 g z$

$$\frac{\partial u_i}{\partial t}+u_j\frac{\partial(u_i)}{\partial x_j}=-\frac{\rho-\rho_0}{\rho_0}g e_{i3}-\frac{1}{\rho_0}\frac{\partial p_m}{\partial x_i}+\nu\frac{\partial^2 u_i}{\partial x_j\partial x_j}$$

For an ideal gas

$$\frac{\partial u_i}{\partial t}+u_j\frac{\partial(u_i)}{\partial x_j}=\frac{T-T_0}{T_0}g e_{i3}-\frac{1}{\rho_0}\frac{\partial p_m}{\partial x_i}+\nu\frac{\partial^2 u_i}{\partial x_j\partial x_j}$$

Buoyancy force per unit mass

II.2. Boundary conditions on fluid-solid interfaces

Solid BCs: no-slip, $u = u_w$

Why no slip?
Is no slip BC always correct?

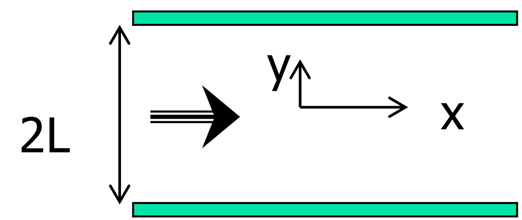
The Navier slip boundary condition

$$u = l_s \frac{\partial u}{\partial y}$$

On the wall that is aligned with the *x* direction

Example, steady-state laminar channel flow

$$0 = g + \nu \frac{\partial^2 u}{\partial y^2}$$



$$u(y = L) = -l_s \frac{\partial u}{\partial y}, \qquad u(y = -L) = l_s \frac{\partial u}{\partial y}$$

l_s is known as the slip length ~ molecular mean free path

The solution

$$u = \frac{gL^2}{2\nu}\left(1 - \frac{y^2}{L^2} + 2\frac{l_s}{L}\right) = u_0\left(1 - \frac{y^2}{L^2} + 2\frac{l_s}{L}\right) \rightarrow \frac{u_{slip}}{u_0} = 2\frac{l_s}{L}$$

Summary

- ✧ The basic fluid mechanics governing equations are reviewed
  - Mass: the continuity equation
  - Momentum: the Navier-Stokes equation
  - Energy: the temperature equation
  - Physical meaning and physical dimension of each variable

- ✧ We will deal with low-speed incompressible viscous flows only.

- ✧ For the problems we will solve in this course, no-slip boundary condition is adequate.